

Chapter 3: In-Class Exercises

Part 1

3.2 Problem: Marguerite receives an allowance of \$20 at the end of each month. Her four-year old brother Stevie receives an allowance of \$1 at month's end. Find the value on January 1 of each of their allowances for the three-month period consisting of January, February, and March, using a monthly effective interest rate of .5%.

3.2 Problem: Candace wishes to buy a new car. Her budget allows her to make monthly payments of \$500, and she has saved \$1,800 for a down payment. Candace qualifies for a 36-month auto loan which has a nominal interest rate of 4.8% convertible monthly. How expensive a car can Candace buy?

3.2

Problem: Seth wishes to borrow \$4,400 so he can pay his tuition. He qualifies for a two-year loan with a monthly effective interest rate of .25% and level payments. Find the amount of Seth's monthly payments under this loan.

3.2

Problem: Mr. Liang wishes to accumulate \$200,000 in a college fund for his newborn daughter. He wishes the money to be available to her on her eighteenth birthday, and he is prepared to make level contributions on her first through eighteenth birthdays. The fund has an annual effective discount rate of 5%. How large will Mr. Liang's contributions need to be?

3.2

Problem: Hyun wishes to contribute \$100 at the end of each month to his savings account until the account has a balance of at least \$3,000, at which time he will buy his grandmother's piano. Hyun's savings account has an annual interest rate of 5.4% convertible monthly. How many months it will take Hyun to accumulate the needed money, and how much money will he then have saved?

3.3

Problem: During 1998, Omar had a dividend payment of \$100 directly deposited to his savings account on the first day of each month. Find the accumulated value of these payments at the end of the year if his savings account had a nominal discount rate of 4.8% payable monthly.

3.3

Problem: Dr. Hillary Street began making contributions to a new retirement account on her thirtieth birthday. She made a contribution of \$4,000 at the beginning of each year through her sixty-fourth birthday. Starting at age sixty-five and continuing through her eightieth birthday, she made a level withdrawal on her birthday. Find the amount of these withdrawals if they completely exhaust the balance in her account, and the annual effective interest rate is 6% until she is sixty-five, then 5% thereafter.

3.4

Problem: Gustave Larson has saved \$20,000. On January 1, he purchases a perpetuity with annual end-of-year payments. The perpetuity price is based on an annual effective interest rate of 5%. What are the annual payments for Gustave's perpetuity?

3.4

Problem: Norman wishes to leave an inheritance to three charities. The total inheritance is a series of level payments at the beginning of each year forever. He wishes charities A and B (which support medical research) to share the payments equally for ten years, and then all payments revert to charity C (that helps needy children). If the shares received by all three charities have the same value at the time of the bequest, find the annual effective interest rate i .

3.5

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her eighteenth birthday using an annual effective interest rate of 5%.

3.5

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her fiftieth birthday using an annual effective interest rate of 5%.

3.5

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her thirtieth birthday using an annual effective interest rate of 5%.

3.5

Problem: Marlene received a gift of a thirty-year annuity-immediate on the day she was born. The annuity pays \$1,000 on her first ten birthdays, \$2,000 on her next ten birthdays, and \$3,000 on the following ten birthdays. (i) If the nominal rate of interest is 5% convertible quarterly, find the value of the annuity on the day she was born. (ii) If the value of the annuity on the day she was born is \$25,000 when calculated using an annual effective rate of interest of j , find j correct to the nearest one-hundredth of a percent.



Problem: Mr. and Mrs. Harper purchase a home for \$256,000. They make a down payment of \$40,000 and finance the remainder of the purchase price with a thirty-year mortgage at an annual effective interest rate of 6.5%. The Harpers sell their home at the end of eight years, just after having made their ninety-sixth end-of-month mortgage payment. The sales price is \$282,000, and the Harpers closing costs are 3% of the selling price. The outstanding loan balance is deducted from the amount the Harpers receive and sent to the lender. How large a check do the Harpers receive, and how much interest did they pay over the eight years?